THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 5: **Measurement of Length and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which instrument is best for measuring the circumference of a tree trunk? 1

(a) divider only **(b)** metre rod **(c)** geometry scale **(d)** measuring tape

2. A scale has its zero end broken. The reading of the object's ends are 2.0 cm and 9.5 cm. The length is: 1

(a) 11.5 cm **(b)** 9.5 cm **(c)** 7.5 cm **(d)** 2.0 cm

3. A ball thrown upward moves in: 1

(a) a zigzag **(b)** a straight line **(c)** a circle **(d)** a loop

4. A plastic ruler heats up and expands. Measurements made with it would be: 1

(a) slightly wrong **(b)** exactly right **(c)** unchanged **(d)** impossible

5. When a student measures the same line three times, she gets 5.1 cm, 5.0 cm, 5.2 cm. The best value is: 1

(a) 5.2 cm **(b)** 5.1 cm **(c)** 5.0 cm **(d)** 15.3 cm

6. A kite flying in a gusty wind shows: 1

(a) random motion **(b)** linear motion **(c)** rotational **(d)** none

7. The thickness of a coin is best measured in: 1

(a) cm only **(b)** mm **(c)** km **(d)** m

8. A wheel has a diameter 70 cm; its circumference as measured by a thread is 220 cm. The ratio is nearly: 1

(a) 3.14 **(b)** 2 **(c)** 5 **(d)** 4

9. A bicycle wheel shows: 1

(a) rest **(b)** rotational and linear (as it moves)

(c) only random **(d)** only linear

10. Which unit is best for the height of Mount Everest? 1

(a) metres **(b)** millimetres **(c)** micrometres **(d)** centimetres

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is the SI unit helpful in science? 2

12. A student reads a length as 7.3 cm from the side. What error is it and how to remove it? 2

13. Why does a passenger in a train feel at rest but a person on the platform says he moves? 2

14. A wheel makes one full turn when it moves 2 m. How many turns for 100 m? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the difference between motion along a straight line and along a curved path with examples. 3

16. Explain, with an example, why the motion of a person can be both at rest and moving. 3

17. State three types of errors that can arise in measuring length. 3

OR

Explain why a bird flying has random motion. 3

18. Classify as periodic or non-periodic: heartbeat, falling leaf, moon revolving, bouncing ball. 3

OR

Explain how a car's movement on a curved road is a combination of motions. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain why ancient units like cubit and foot were replaced by standard units. Describe the SI system. 5

OR

Explain the parallax error with a sketch described in words and suggest ways to avoid it. 5

20. A cyclist rides 3.6 km to school. His wheel has a circumference of 1.8 m. Calculate how many turns the wheel makes. 5

OR

Discuss the types of motion of a bicycle: wheels, pedals, handle, and the cycle as a whole. 5

END OF QUESTION PAPER — **Total: 40 Marks**

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